

Natural Deduction

in Predicate Logic

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- ① Direct Proof
 - Rules of inference
 - Change of Quantifier Rules
- ② Conditional Indirect Proof

Rules of inference

1. Universal instantiation (UI):

Allowed:

$$\frac{\forall x F(x)}{F(y)}$$

$F(x)$: Statement function. Examples: $[p(x) \Rightarrow q(x)]$, $[p(x) \wedge q(x)]$, etc

x, y : variables

a : a constant

Allowed:

$$\frac{\forall x F(x)}{F(a)}$$

2. Universal Generalization (UG):

Allowed:

$$\frac{F(y)}{\forall x F(x)}$$

$F(x)$: Statement function. Examples: $[p(x) \Rightarrow q(x)]$, $[p(x) \wedge q(x)]$, etc

x, y : variables

a : a constant

NOT Allowed:

$$\frac{F(a)}{\forall x F(x)}$$

Rules of inference

3. Existential instantiation (EI):

Allowed:

$$\frac{\exists x F(x)}{F(a)}$$

NOT Allowed:

$$\frac{\exists x F(x)}{F(y)}$$

$F(x)$: Statement function. Examples: $[p(x) \Rightarrow q(x)]$, $[p(x) \wedge q(x)]$, etc

x, y : variables

a : a constant

a : The existential name a must be a new name that has not occurred in any previous line.

4. Existential Generalization (EG):

Allowed:

$$\frac{F(a)}{\exists x F(x)}$$

Allowed:

$$\frac{F(y)}{\exists x F(x)}$$

$F(x)$: Statement function. Examples: $[p(x) \Rightarrow q(x)]$, $[p(x) \wedge q(x)]$, etc

x, y : variables

a : a constant

Universal Instantiation's Application

No.	Argument
1.	All economists are social scientists.
2.	Milton Friedman is an economist.
3.	Therefore, Milton Friedman is a social scientist.

Table : List of Predicates

No.	Predicates	Meaning
1.	$economist(x)$	x is an economist.
2.	$scientist(x)$	x is social scientist.

Table : Translation to Predicate Logic

No.	English	Predicate Logic
1.	All economists are social scientists.	$\forall x[\text{economist}(x) \Rightarrow \text{scientist}(x)]$
2.	Milton Friedman is an economist.	$\text{economist}("M.Friedman")$
3.	Milton Friedman is a social scientist.	$\text{scientist}("M.Friedman")$

Table : Proof

No.	Predicates	Notes
1.	$\forall x[\text{economist}(x) \Rightarrow \text{scientist}(x)]$	Premise
2.	$\text{economist}("M.Friedman")$ $\text{scientist}("M.Friedman")$	Premise Expected Conclusion
3.	$\text{economist}("M.Friedman") \Rightarrow \text{scientist}("M.Friedman")$	1, UI
4.	$\text{scientist}("M.Friedman")$ ■	2, 3, MP Proved

Universal Generalization's Application

No.	Argument
1.	All psychiatrists are doctors.
2.	All doctors are college graduates.
3.	Therefore, all psychiatrists are college graduates.

Table : List of Predicates

No.	Predicates	Meaning
1.	<i>psychiatrist</i> (x)	x is a psychiatrist.
2.	<i>doctor</i> (x)	x is a doctor.
3.	<i>graduate</i> (x)	x is a college graduate.

Table : Translation to Predicate Logic

No.	English	Predicate Logic
1.	All psychiatrists are doctors.	$\forall x[\text{psychiatrist}(x) \Rightarrow \text{doctor}(x)]$
2.	All doctors are college graduates.	$\forall x[\text{doctor}(x) \Rightarrow \text{graduate}(x)]$
3.	All psychiatrists are college graduates.	$\forall x[\text{psychiatrist}(x) \Rightarrow \text{graduate}(x)]$

Table : Proof

No.	Predicates	Notes
1.	$\forall x[\text{psychiatrist}(x) \Rightarrow \text{doctor}(x)]$	Premise
2.	$\forall x[\text{doctor}(x) \Rightarrow \text{graduate}(x)]$	Premise
	$\forall x[\text{doctor}(x) \Rightarrow \text{graduate}(x)]$	Expected Conclusion
3.	$\text{psychiatrist}(y) \Rightarrow \text{doctor}(y)$	1, UI
4.	$\text{doctor}(y) \Rightarrow \text{graduate}(y)$	2, UI
5.	$\text{psychiatrist}(y) \Rightarrow \text{graduate}(y)$	3, 4, HS
6.	$\forall x[\text{psychiatrist}(x) \Rightarrow \text{graduate}(x)]$	5, UG
	■	Proved

Existential Generalization's Application

No.	Argument
1.	All tenors are singers.
2.	Placido Domingo is a tenor.
3.	Therefore, there is at least one singer.

Table : List of Predicates

No.	Predicates	Meaning
1.	$tenor(x)$	x is a tenor.
2.	$singer(x)$	x is a singer.

Rules of inference

Table : Translation to Predicate Logic

No.	English	Predicate Logic
1.	All tenors are singers.	$\forall x[tenor(x) \Rightarrow singer(x)]$
2.	Placido Domingo is a tenor.	$tenor("P.Domingo")$
3.	There is at least one singer.	$\exists x singer(x)$

Table : Proof

No.	Predicates	Notes
1.	$\forall x[tenor(x) \Rightarrow singer(x)]$	Premise
2.	$tenor("P.Domingo")$	Premise
	$\exists x singer(x)$	Expected Conclusion
3.	$tenor("P.Domingo") \Rightarrow singer("P.Domingo")$	1, UI
4.	$singer("P.Domingo")$	2, 3, MP
5.	$\exists x singer(x)$	4, EG
	■	Proved

Existential Instantiation's Application

No.	Argument
1.	All attorneys are college graduates.
2.	Some attorneys are golfers.
3.	Therefore, some golfers are college graduates.

Table : List of Predicates

No.	Predicates	Meaning
1.	<i>attorney</i> (x)	x is an attorney.
2.	<i>graduate</i> (x)	x is a college graduate.
2.	<i>golfer</i> (x)	x is a golfer.

Table : Translation to Predicate Logic

No.	English	Predicate Logic
1.	All attorneys are college graduates.	$\forall x[\text{attorney}(x) \Rightarrow \text{graduate}(x)]$
2.	Some attorneys are golfers.	$\exists x[\text{attorney}(x) \wedge \text{golfer}(x)]$
3.	some golfers are college graduates.	$\exists x[\text{golfer}(x) \wedge \text{graduate}(x)]$

Table : Proof

No.	Predicates	Notes
1.	$\forall x[\text{attorney}(x) \Rightarrow \text{graduate}(x)]$	Premise
2.	$\exists x[\text{attorney}(x) \wedge \text{golfer}(x)]$	Premise
	$\exists x[\text{golfer}(x) \wedge \text{graduate}(x)]$	Expected Conclusion
3.	$\text{attorney}(a) \wedge \text{golfer}(a)$	2, EI, a : a constant
4.	$\text{attorney}(a) \Rightarrow \text{graduate}(a)$	1, UI
5.	$\text{attorney}(a)$	3, Simplification
6.	$\text{golfer}(a)$	3, Simplification
7.	$\text{graduate}(a)$	4, 5, MP
8.	$\text{golfer}(a) \wedge \text{graduate}(a)$	6, 7, Conjunction
9.	$\exists x[\text{golfer}(x) \wedge \text{graduate}(x)]$	8, EG
	■	Proved

Change of Quantifier Rules

Change of Quantifier Rules (CQ)

$$\forall x F(x) \Leftrightarrow \neg \exists x \neg F(x)$$

$$\exists x F(x) \Leftrightarrow \neg \forall x \neg F(x)$$

$$\neg \forall x F(x) \Leftrightarrow \exists x \neg F(x)$$

$$\neg \exists x F(x) \Leftrightarrow \forall x \neg F(x)$$

$F(x)$: Statement function. Examples: $[p(x) \Rightarrow q(x)]$, $[p(x) \wedge q(x)]$, etc

x : variable

Change of Quantifier Rules

Table : CQ: Example

No.	Predicates	Notes
1.	$\neg\exists x[p(x) \wedge \neg q(x)]$	Premise
2.	$\neg\forall x[\neg r(x) \vee q(x)]$	Premise
	$\exists x\neg p(x)$	Expected Conclusion
3.	$\forall x\neg[p(x) \wedge \neg q(x)]$	1, CQ
4.	$\exists x\neg[\neg r(x) \vee q(x)]$	2, CQ
5.	$\neg[\neg r(a) \vee q(a)]$	4, EI
6.	$\neg[p(a) \wedge \neg q(a)]$	3, UI
7.	$r(a) \wedge \neg q(a)$	5, DM, DN
8.	$\neg p(a) \vee q(a)$	6, DM, DN
9.	$\neg q(a)$	7, Simplification
10.	$\neg p(a)$	8, 9, DS
11.	$\exists x\neg p(x)$	10, EG
	■	Proved

Conditional and Indirect Proof

UG: Important Note

- UG: $\frac{F(y)}{\forall x F(x)}$
- UG must not be used within the scope of an indented sequence if the instantial variable occurs free in the first line of that sequence.

Table : An invalid reasoning

No.	Predicates	Notes
1.	$\forall x[r(x) \Rightarrow \forall x s(x)]$	Premise
	$\forall x[r(x) \Rightarrow s(x)]$	Expected Conclusion
2.	$r(x)$	ACP
3.	$\forall x r(x)$	2, UG (invalid)
4.	$\forall x s(x)$	1, 3, MP
5.	$s(x)$	4, UI
6.	$r(x) \Rightarrow s(x)$	2-5, CP
7.	$\forall x[r(x) \Rightarrow s(x)]$	2-5, CP

Proved



Invalid reasoning, why?

- $r(x)$: means "x is a rabbit."
- $s(x)$: means "x is a snake."
- Premise: If everything in the universe is a rabbit, then everything in the universe is a snake.
- Premise is true because its antecedent is false.
- Conclusion: All rabbits are snakes.
- Conclusion is false because rabbit is not snake.
- The above reasoning is false.

Conditional and Indirect Proof

UG: Important Note

- UG: $\frac{F(y)}{\forall x F(x)}$
- UG must not be used within the scope of an indented sequence if the instancial variable occurs free in the first line of that sequence.

Table : A correct reasoning

No.	Predicates	Notes
1.	$\forall x[r(x) \Rightarrow s(x)]$	Premise
	$\forall x r(x) \Rightarrow \forall x s(x)$	Expected Conclusion
2.	$\forall x r(x)$	ACP
3.	$r(x)$	2, UI
4.	$r(x) \Rightarrow s(x)$	1, UI
5.	$s(x)$	3, 4, MP
6.	$\forall x s(x)$	5, UG
7.	$\forall x r(x) \Rightarrow \forall x s(x)$	2-6, CP

Proved

